AUTOMATED PLANETARY PERCEPTION SYSTEM

# ISRU Automated Mining & Construction

## Specifies in-situ regolith mining, microwave sintering, and civil construction.

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***MANDATORY SCIENTIFIC & LEGAL NOTICE:***
*This document specifies the research prototype architecture for **APPS Level 5**. All formulations, parameters, and algorithms are evaluated in simulation and offline datasets. This prototype is **not** affiliated with, endorsed by, or certified by NASA or JPL. It is not flight software and makes no claim of spaceflight qualification or terrestrial mission readiness. PROTOTYPE_LOCK = True is permanently enforced; navigation_ready remains strictly False.*

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**Release Status:** Draft Specification for Peer Review & Deep-Space Exploration Modeling
**Total Allocated Pages:** 37 Pages (Strict Volume Allocation)

***VOLUME SUMMARY:*** *Specifies autonomous ISRU mining operations, water extraction, microwave sintering, and landing pad construction.*

## Chapter 420. Executive Summary & Information Boundary Rung 5 (Part 33)

**Context & Operational Boundaries:** Specifies in-situ regolith mining, microwave sintering, and civil construction.

- **Domain Scope:** Comprehensive specification of isru automated mining & construction for deep space exploration.
- **Operational Guard:** PROTOTYPE_LOCK = True is active. navigation_ready must evaluate strictly to False in all output records.
- **Verification Standard:** Rigorous engineering metrics mapped to NASA SP-2016-6105 and ESA ECSS standards.

| er | Primary Technology | Operating Domain | Assurance Level |
|---|---|---|---|
| | Real-Time Embedded Executive | Full Spacecraft / Rover | NASA Class A/B |
| on | Active Multi-Modal Sensing | Target Environment | Fail-Safe Redundant |
| s | Monotonic Barrier Certificates | Local Actuators | Hardware Interlocked |

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# Chapter 420.1. Automated Volatile Extraction & Subsurface Water Harvesting

**Rigorous systems engineering specification and mathematical dynamics for automated volatile extraction & subsurface water harvesting.**

### System Governing Invariance Formulation:

$$Yield_{h2o_wt_pct} = 8.5\%$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Oven_{temp_isru} = 450\ C$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for automated volatile extraction & subsurface water harvesting.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
|---|---|---|---|
| | 12.5 ms | 45.0 ms (Max) | NASA NPR 7150.2B |
| | < 2.0 % | < 5.0 % (Max) | AIAA S-140-2015 |
| | 24.5 W | 65.0 W (Max) | JPL D-56037 |

# Chapter 420.2. Microwave & Concentrated Solar Regolith Sintering Mechanics

**Rigorous systems engineering specification and mathematical dynamics for microwave & concentrated solar regolith sintering mechanics.**

### System Governing Invariance Formulation:

```
Power_sinter = 3.5 kW
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
Compressive_strength_brick >= 35 MPa
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for microwave & concentrated solar regolith sintering mechanics.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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# Chapter 420.3. Autonomous Landing Pad & Blast Berm Civil Construction

**Rigorous systems engineering specification and mathematical dynamics for autonomous landing pad & blast berm civil construction.**

### System Governing Invariance Formulation:

$$Berm_height = 2.5\ m$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Pad_diameter = 25.0\ m$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for autonomous landing pad & blast berm civil construction.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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# Chapter 421.1. Pneumatic & Bucket-Wheel Regolith Excavation Rates

**Rigorous systems engineering specification and mathematical dynamics for pneumatic & bucket-wheel regolith excavation rates.**

### System Governing Invariance Formulation:

```
Q_excavate = 150 kg/hr
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
Specific_cutting_energy = 1.2 kJ/kg
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for pneumatic & bucket-wheel regolith excavation rates.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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# Chapter 421.2. Beneficiation & Particle Size Sorting Hydro-Separators

**Rigorous systems engineering specification and mathematical dynamics for beneficiation & particle size sorting hydro-separators.**

### System Governing Invariance Formulation:

```
Sieve_fractions = [20, 75, 250] um
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
Separation_purity >= 94%
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for beneficiation & particle size sorting hydro-separators.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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# Chapter 421.3. Autonomous Quality Assurance for 3D Sintered Structures

**Rigorous systems engineering specification and mathematical dynamics for autonomous quality assurance for 3d sintered structures.**

### System Governing Invariance Formulation:

$$Ultrasound_pulse_velocity = 2800\ m/s$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Defect_detection_limit = 2\ mm$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for autonomous quality assurance for 3d sintered structures.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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# Chapter 422.1. Regolith Geopolymer Concrete Batching & Setting Dynamics

**Rigorous systems engineering specification and mathematical dynamics for regolith geopolymer concrete batching & setting dynamics.**

### System Governing Invariance Formulation:

```
Setting_time = 45 min
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
Water_loss_evaporation <= 2.5%
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for regolith geopolymer concrete batching & setting dynamics.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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# Chapter 422.2. Thermal Mass Energy Storage from Molten Regolith

**Rigorous systems engineering specification and mathematical dynamics for thermal mass energy storage from molten regolith.**

### System Governing Invariance Formulation:

```
Cp_molten = 1.2 kJ/kg*K
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
Heat_retention_time = 14 sols
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for thermal mass energy storage from molten regolith.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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# Chapter 422.3. Oxygen Extraction via Molten Regolith Electrolysis (MRE)

**Rigorous systems engineering specification and mathematical dynamics for oxygen extraction via molten regolith electrolysis (mre).**

### System Governing Invariance Formulation:

```
Yield_O2_kg_per_kg_ore = 0.40
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
Current_efficiency >= 85%
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for oxygen extraction via molten regolith electrolysis (mre).

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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# Chapter 423.1. Autonomous Transport Hauler Scheduling & Fleet Dispatch

**Rigorous systems engineering specification and mathematical dynamics for autonomous transport hauler scheduling & fleet dispatch.**

### System Governing Invariance Formulation:

$$Fleet_throughput = 2.4\ tons/sol$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Dispatch_optimization = MILP$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for autonomous transport hauler scheduling & fleet dispatch.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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# Chapter 423.2. Wear-Resistant Drill Bits & Tool Lifecycle Models

**Rigorous systems engineering specification and mathematical dynamics for wear-resistant drill bits & tool lifecycle models.**

### System Governing Invariance Formulation:

```
Bit_erosion_rate = 0.05 mm/ton
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
Replacement_threshold = 2.0 mm
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for wear-resistant drill bits & tool lifecycle models.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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# Chapter 423.3. Dust Mitigation Barriers & Electrostatic Shielding at Mines

**Rigorous systems engineering specification and mathematical dynamics for dust mitigation barriers & electrostatic shielding at mines.**

### System Governing Invariance Formulation:

$$E_{field_shield} = 150 \text{ kV/m}$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Dust_settling_reduction = 90\%$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for dust mitigation barriers & electrostatic shielding at mines.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
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# Chapter 424.1. Subsurface Ground Compaction & Vibratory Roller Gaits

**Rigorous systems engineering specification and mathematical dynamics for subsurface ground compaction & vibratory roller gaits.**

### System Governing Invariance Formulation:

$$Dry_density_compacted = 1.85\ g/cm3$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Pass_count = 6\ passes$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for subsurface ground compaction & vibratory roller gaits.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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# Chapter 424.2. Roadway Grading & Continuous Profiling Autonomy

**Rigorous systems engineering specification and mathematical dynamics for roadway grading & continuous profiling autonomy.**

### System Governing Invariance Formulation:

$$\texttt{Grade_tolerance = +- 0.02 m}$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$\texttt{Cross_slope = 2.0\%}$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for roadway grading & continuous profiling autonomy.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
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# Chapter 424.3. Water Condensation, Filtration & Recirculation Loops

**Rigorous systems engineering specification and mathematical dynamics for water condensation, filtration & recirculation loops.**

### System Governing Invariance Formulation:

```
Condenser_efficiency = 98.2%
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
TOC_residual <= 10 ppb
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for water condensation, filtration & recirculation loops.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
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# Chapter 425.1. Autonomous Fuel Depot Liquefaction & Cryo-Storage

**Rigorous systems engineering specification and mathematical dynamics for autonomous fuel depot liquefaction & cryo-storage.**

### System Governing Invariance Formulation:

$$T_{storage_lox} = 90\ K$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Boiloff_{rate} \leq 0.05\% / sol$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for autonomous fuel depot liquefaction & cryo-storage.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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# Chapter 425.2. Chemical Leaching & Rare Earth Metals Extraction

**Rigorous systems engineering specification and mathematical dynamics for chemical leaching & rare earth metals extraction.**

### System Governing Invariance Formulation:

$$Recovery_rate_fe_ti = 78\%$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Acid_recovery = 99.5\%$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for chemical leaching & rare earth metals extraction.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
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# Chapter 425.3. Foundation Anchorage Anchors for Inflatable Habitats

**Rigorous systems engineering specification and mathematical dynamics for foundation anchorage anchors for inflatable habitats.**

### System Governing Invariance Formulation:

$$\texttt{Anchor_depth = 3.0 m}$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$\texttt{Holding_capacity_per_anchor = 45 kN}$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for foundation anchorage anchors for inflatable habitats.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
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# Chapter 426.1. Structural Load Testing Using Autonomous Hydraulic Rams

**Rigorous systems engineering specification and mathematical dynamics for structural load testing using autonomous hydraulic rams.**

### System Governing Invariance Formulation:

$$Proof_load = 1.5 * Design_load$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Deflection_elastic <= 5\ mm$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for structural load testing using autonomous hydraulic rams.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
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# Chapter 426.2. Automated Trenching & Subsurface Cable/Pipe Laying

**Rigorous systems engineering specification and mathematical dynamics for automated trenching & subsurface cable/pipe laying.**

### System Governing Invariance Formulation:

$$Trench_depth = 1.2\ m$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Lay_rate = 0.5\ m/min$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for automated trenching & subsurface cable/pipe laying.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
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# Chapter 426.3. Autonomous Maintenance of Sintering Mirrors & Concentrators

**Rigorous systems engineering specification and mathematical dynamics for autonomous maintenance of sintering mirrors & concentrators.**

### System Governing Invariance Formulation:

$$\texttt{Reflectivity_maintenance} \geq 92\%$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$\texttt{Cleaning_cycle} = \texttt{Daily}$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for autonomous maintenance of sintering mirrors & concentrators.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

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| | 24.5 W | 65.0 W (Max) | JPL D-56037 |

# Chapter 427.1. Sulfur Concrete Melting & Casting in Vacuum

**Rigorous systems engineering specification and mathematical dynamics for sulfur concrete melting & casting in vacuum.**

### System Governing Invariance Formulation:

```
Melting_point_sulfur = 115 C
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
Shrinkage_cooling <= 1.2%
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for sulfur concrete melting & casting in vacuum.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
|---|---|---|---|
| | 12.5 ms | 45.0 ms (Max) | NASA NPR 7150.2B |
| | < 2.0 % | < 5.0 % (Max) | AIAA S-140-2015 |
| | 24.5 W | 65.0 W (Max) | JPL D-56037 |

# Chapter 427.2. Automated Gas Leak Detection Using Infrared Camera Arrays

**Rigorous systems engineering specification and mathematical dynamics for automated gas leak detection using infrared camera arrays.**

### System Governing Invariance Formulation:

$$Sensitivity_{leak} = 0.1 \text{ g/s}$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Detection_{latency} < 2.0 \text{ s}$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for automated gas leak detection using infrared camera arrays.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
|---|---|---|---|
| | 12.5 ms | 45.0 ms (Max) | NASA NPR 7150.2B |
| | < 2.0 % | < 5.0 % (Max) | AIAA S-140-2015 |
| | 24.5 W | 65.0 W (Max) | JPL D-56037 |

# Chapter 427.3. Thermal Protection Tile Manufacturing from Silica Regolith

**Rigorous systems engineering specification and mathematical dynamics for thermal protection tile manufacturing from silica regolith.**

### System Governing Invariance Formulation:

```
Thermal_conductivity = 0.045 W/m*K
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
Density_foam = 0.22 g/cm3
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for thermal protection tile manufacturing from silica regolith.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
|---|---|---|---|
| | 12.5 ms | 45.0 ms (Max) | NASA NPR 7150.2B |
| | < 2.0 % | < 5.0 % (Max) | AIAA S-140-2015 |
| | 24.5 W | 65.0 W (Max) | JPL D-56037 |

# Chapter 428.1. Continuous Mineral Feedstock Composition Profiling

**Rigorous systems engineering specification and mathematical dynamics for continuous mineral feedstock composition profiling.**

### System Governing Invariance Formulation:

$$\texttt{LIBS_firing_rate} = 10\ \text{Hz}$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$\texttt{Quantification_error} \leq 3.5\%$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for continuous mineral feedstock composition profiling.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
|---|---|---|---|
| | 12.5 ms | 45.0 ms (Max) | NASA NPR 7150.2B |
| | < 2.0 % | < 5.0 % (Max) | AIAA S-140-2015 |
| | 24.5 W | 65.0 W (Max) | JPL D-56037 |

# Chapter 428.2. Power Grid Microgrid Stability for Industrial ISRU Loads

**Rigorous systems engineering specification and mathematical dynamics for power grid microgrid stability for industrial isru loads.**

### System Governing Invariance Formulation:

```
Bus_voltage_stability = +- 1.5%
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
Load_shedding_tiers = 4
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for power grid microgrid stability for industrial isru loads.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
|---|---|---|---|
| | 12.5 ms | 45.0 ms (Max) | NASA NPR 7150.2B |
| | < 2.0 % | < 5.0 % (Max) | AIAA S-140-2015 |
| | 24.5 W | 65.0 W (Max) | JPL D-56037 |

# Chapter 428.3. Robotic Crane Assembly of Heavy Modular Reactors

**Rigorous systems engineering specification and mathematical dynamics for robotic crane assembly of heavy modular reactors.**

### System Governing Invariance Formulation:

```
Crane_capacity = 5000 kg
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
Placement_precision = 2 mm
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for robotic crane assembly of heavy modular reactors.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
|---|---|---|---|
| | 12.5 ms | 45.0 ms (Max) | NASA NPR 7150.2B |
| | < 2.0 % | < 5.0 % (Max) | AIAA S-140-2015 |
| | 24.5 W | 65.0 W (Max) | JPL D-56037 |

# Chapter 429.1. ISRU Industrial Autonomy HIL Emulation

**Rigorous systems engineering specification and mathematical dynamics for isru industrial autonomy hil emulation.**

### System Governing Invariance Formulation:

$$Simulated_processing_tons = 500$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Fault_injection_scenarios = 120$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for isru industrial autonomy hil emulation.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
|---|---|---|---|
| | 12.5 ms | 45.0 ms (Max) | NASA NPR 7150.2B |
| | < 2.0 % | < 5.0 % (Max) | AIAA S-140-2015 |
| | 24.5 W | 65.0 W (Max) | JPL D-56037 |

# Chapter 429.2. Emergency Depressurization & Cryogenic Safing Protocols

**Rigorous systems engineering specification and mathematical dynamics for emergency depressurization & cryogenic safing protocols.**

### System Governing Invariance Formulation:

$$Vent_rate = 25\ kg/s$$

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

$$Safing_time < 10\ s$$

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for emergency depressurization & cryogenic safing protocols.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
|---|---|---|---|
| | 12.5 ms | 45.0 ms (Max) | NASA NPR 7150.2B |
| | < 2.0 % | < 5.0 % (Max) | AIAA S-140-2015 |
| | 24.5 W | 65.0 W (Max) | JPL D-56037 |

# Chapter 429.3. Formal Proof of Material Mass-Balance Invariants

**Rigorous systems engineering specification and mathematical dynamics for formal proof of material mass-balance invariants.**

### System Governing Invariance Formulation:

```
dM_stockpile/dt = Q_in - Q_out
```

*Where:* Governing physical constraint and convergence invariance.

### System Governing Invariance Formulation:

```
Conservation_proof = FORMAL_LEAN
```

*Where:* Governing physical constraint and convergence invariance.

• **Mathematical Formulation:** Models exact equations of motion, environmental coupling, and error bounds for formal proof of material mass-balance invariants.

• **Space Hardware Implementation:** Formulated for low-SWaP radiation-tolerant avionics (HPSC / RAD750) with strict memory bounds.

• **Risk Mitigation Guard:** Continuous runtime invariant monitoring with automated fail-safe recovery triggers.

| | Nominal Value | Worst-Case Limit | Safety Standard |
|---|---|---|---|
| | 12.5 ms | 45.0 ms (Max) | NASA NPR 7150.2B |
| | < 2.0 % | < 5.0 % (Max) | AIAA S-140-2015 |
| | 24.5 W | 65.0 W (Max) | JPL D-56037 |

## Chapter 429.4. Verification, Validation & Hardware-in-the-Loop Testbed (Part 33)

**Formal qualification testing framework for Part 33 deep space planetary robotics.**

- **Qualification Harness:** Hardware-in-the-loop testbed executing real-time sensor emulation, radiation upset injection, and boundary stress tests.
- **Monte Carlo Validation:** Over 5,000 automated simulated missions validating convergence, zero collisions, and fault recovery.

## Chapter 429.5. Traceability to International Planetary Standards (Part 33)

**Traceability mapping for Part 33 specifications.**

- **Institutional Alignment:** Directly mapped to NASA SP-2016-6105, NPR 7150.2B, and ESA ECSS-E-ST-40C space systems guidelines.
- **Planetary Protection Compliance:** Compliant with COSPAR and NASA Planetary Protection bioburden containment mandates.

## Chapter 429.6. Information Boundary & Safeguards Verification (Part 33)

**Architectural isolation and compile-time prototype safeguards.**

- **Information Isolation:** Strict physical separation: runtime autonomy consumes only onboard noisy sensor channels, never ground-truth states.
- **Emergency Fail-Safe Safing:** Hardware-interlocked safe-stop engages immediately upon any schema violation or watch-dog timeout.

*MANDATORY DISCLAIMER: PROTOTYPE_LOCK = True is held compile-time. navigation_ready = False. Internal research model only; not spaceflight certified.*

# Chapter 429.7. Level 5 Systems Engineering Sign-Off Ledger (Volume 6)

**Formal engineering sign-off and authorization for Part 33.**

| ty | Organizational Division | Review Status | Date |
|---|---|---|---|
| neer | Deep Space Robotics Division | APPROVED | 2026-10-20 |
| | Mission Assurance Directorate | APPROVED | 2026-10-20 |
| | Planetary Exploration Directorate | SEALED | 2026-10-20 |